

Ideological discrepancy between Facebook pages and shared news is associated with reaction homogeneity

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Abstract

Cross-cutting political content is often expected to mitigate polarization, yet encounters across ideological lines do not necessarily foster more constructive engagement. We analyzed 4,584 posts from 1,243 Facebook pages sharing 761 political news links in a dataset centered on Brazil’s 2018 presidential election, and introduced Bias Discrepancy, which measures how far a shared link lies from the page’s ideological sharing repertoire. We examined how Bias Discrepancy related to Content Reaction Positivity (how positively the same link was received in other posts), the Reaction Homogeneity Index (whether a post’s own reactions were one-sided or mixed), and the toxicity of the page’s accompanying text. Reaction homogeneity was lowest at both extremes of Bias Discrepancy, and posts furthest from the page’s repertoire had the highest average toxicity. In a multivariable model, greater Bias Discrepancy was associated with lower reaction homogeneity, with a magnitude similar to the link’s content bias and toxicity and smaller than Content Reaction Positivity. When a page shared news that departed from its repertoire, its accompanying text tended to be more toxic and its audience’s reactions more mixed.

Introduction

The rise of social media as a primary source of political information has transformed news consumption and how individuals engage with political content [1]. On Facebook, both ranking algorithms and individual choices shape exposure to political content. In a field experiment that randomized subscriptions to news outlets, Facebook’s ranking supplied fewer posts from counter-attitudinal outlets even to users who had subscribed to them [1], while across the platform as a whole individual choices did more than ranking to limit cross-cutting exposure [2]. By curating personalized content based on users’ past interactions, platforms may in turn contribute to the formation of “filter bubbles,” informational ecosystems in which exposure to divergent views is limited and pre-existing beliefs are reinforced [3, 4]. This process has been associated with increased polarization, as homophilous networks [5] promote the formation of ideologically homogeneous groups where misinformation can spread quickly [6].

These structural conditions also have affective consequences. Because echo chambers amplify political beliefs while limiting exposure to opposing viewpoints [3, 7], hostility between opposing groups may intensify [8], and interactions may become increasingly toxic [9]. Elections are one setting where these dynamics play out, since social media is used for voter mobilization and for building campaign narratives. Brazil’s 2018 presidential election is one example. In that election, Jair Bolsonaro built his campaign

around an anti-corruption agenda and used social media as his main channel to communicate with voters [10], and retweet networks during the election showed tight clustering around ideological poles [11]. These dynamics reflected an intensification of affective polarization, more pronounced toward candidates than toward parties, and accompanied the growth and radicalization of the political right [12]. Later analyses of the 2022 election indicate that polarization continued to increase [13].

Most existing work on these dynamics focuses on network exposure [2, 14], content diffusion [6, 7], or structural polarization metrics [15, 16]. By comparison, less attention has been given to what happens when a Facebook page shares political news that departs from its ideological sharing repertoire. Such departures matter because cross-cutting content does not necessarily produce more constructive engagement, and prior work shows that encounters across ideological boundaries can be accompanied by negative affect and nonconstructive interaction [8, 17]. Studying them at the level of individual Facebook posts makes it possible to connect ideological discrepancy to the reactions and language of a particular post.

We address this gap using Brazil’s 2018 presidential election as a case study and ask two research questions. (RQ1) How is the ideological discrepancy between a shared news link and a page’s ideological sharing repertoire related to audience reactions and to the toxicity of the accompanying post text? (RQ2) How does the association between this ideological discrepancy and the homogeneity of audience reactions compare with associations involving other page-, content-, and post-level characteristics?

The posts we analyze come from public Facebook pages. We call these accounts *pages* throughout, rather than publishers, because most of them are not news organizations. They are public Facebook pages and verified accounts, and the term keeps them distinct from the *audience* whose reactions we measure. For each page, we characterize its ideological sharing repertoire with a bias score (b_p) estimated from the ideologically labeled news links it has shared, and define Bias Discrepancy (Δ_b) as the absolute difference between this score and the bias score of a particular shared link (b_c). A high Δ_b identifies a link that departs from that repertoire. It does not, however, establish the page’s independently held ideology or its reason for sharing that link. Bias Discrepancy is the measure we use throughout for the ideological discrepancy in the two research questions, and we relate it to how audiences respond and to the language pages use when sharing. Audience response is captured by two measures: Content Reaction Positivity (*CRP*), the positivity of reactions to other posts sharing the same link, and the Reaction Homogeneity Index (*RHI*), which records whether a post’s reactions fall predominantly on one side or are divided between them. Toxicity (*Tox*), by contrast, characterizes the post rather than the audience, quantifying hostile language in the text written by the page alongside the link. The linked articles are further characterized through zero-shot topic classification. Finally, to address RQ2, we model *RHI* using a Generalized Linear Mixed Model (GLMM) with a Beta distribution, estimating how Bias Discrepancy, Content Reaction Positivity, toxicity, and content topics relate to reaction homogeneity while accounting for differences across pages and shared news links.

This framework yields two main contributions. First, we connect departures from a Facebook page’s ideological sharing repertoire to audience reactions and to the toxicity of the accompanying post text. We found that reaction homogeneity was lowest at both ends of the Bias Discrepancy distribution, that posts in the highest Bias Discrepancy quartile had the highest average toxicity, and that Content Reaction Positivity was lowest in the lowest quartile. Second, we use a multivariable mixed model to compare how Bias Discrepancy, Content Reaction Positivity, toxicity, and other predictors relate to reaction homogeneity. In that model, Content Reaction Positivity had the strongest association, followed at similar magnitudes by Bias Discrepancy, content bias, and

toxicity, and the negative association between Bias Discrepancy and reaction homogeneity persisted across the sensitivity analyses.

The remainder of this paper is structured as follows: Section ‘Related work’ reviews the relevant literature. Section ‘Problem Statement’ defines the research problem. Section ‘Methodology’ describes the dataset and analytical methods. Section ‘Results’ presents the empirical findings, and Section ‘Conclusion’ summarizes the main findings, their implications, and limitations. Finally, Section ‘Ethical considerations’ addresses the ethical and privacy aspects of the study.

Related work

Research in computational social science has examined how user behavior, social ties, and algorithmic systems contribute to ideological echo chambers [3, 18] and political homophily [5, 19, 20], across multiple platforms and methods. More recent work reports that social media platforms shape perceptions of polarization [21], that social sorting reinforces affective polarization across digital interactions [22], and that partisanship can either bridge or deepen online echo chambers [23]. Within this literature, three lines of research inform the present study: (i) how exposure to cross-cutting content is shaped and what follows from it, (ii) how existing polarization metrics capture ideological divisions in online networks, and (iii) how affective responses and emotional reactions structure engagement with political content.

Research on algorithmic curation examines how recommendation systems shape users’ exposure to ideologically diverse content, and how much of that shaping is algorithmic rather than chosen. For example, [2] found that individual choices, more than algorithmic ranking, limit cross-cutting content on Facebook, while [14] showed that exposure to untrustworthy websites during the 2016 U.S. election was concentrated among a subset of users with strong preferences for pro-attitudinal information. Beyond Facebook, personalization on YouTube has been associated with the reinforcement of echo chambers [24]. Separately, a Twitter field experiment found that exposure to opposing political views can increase polarization among some users [8]. By contrast, a field experiment on Facebook found the opposite direction for affective attitudes, with exposure to counter-attitudinal outlets reducing negative feeling toward the opposing party [1]. Studies of cross-partisan interaction add a further qualification. While [17] observed significant engagement among users with differing political views on Twitter, these interactions were often characterized by negative, nonconstructive stances, suggesting that breaking echo chambers does not necessarily lead to constructive dialogue. Similarly, examining downstream effects on Reddit, [25] showed that receiving a cross-party reply rarely increases subsequent engagement with out-party communities, instead reinforcing in-party participation. These findings suggest that exposure patterns alone do not explain how polarization appears in online interactions.

A second line of research focuses on measuring polarization in online networks. Metrics such as $RP(H)$ [26] and controversy frameworks [15, 27] have been used to quantify ideological divisions in retweet networks. These approaches describe the structural organization of polarized communication, particularly in diffusion-based platforms such as Twitter. However, they primarily capture patterns of information flow and network structure, making them less suited to measuring engagement dynamics on platforms like Facebook, where interaction often occurs through reactions and comments rather than content resharing. For this reason, we use an $RP(H)$ -derived score only to characterize news links, and we measure audience response on Facebook through reactions.

A third line of research examines the role of affective dynamics in online political communication. Prior work links textual features to affective polarization, examining

how sentiment, emotion, and misinformation structure political discourse online. For instance, political alignment can often be inferred directly from language use and framing [28], and false news has been found to spread farther and faster than true news, with responses to false stories more often expressing fear, disgust, and surprise [7]. Platform-specific emotional dynamics further influence engagement patterns. For example, sadness has been shown to play a prominent role in the sharing of political content on Facebook [29], reinforcing belief alignment within ideologically homogeneous networks [6]. These affective asymmetries may also be amplified by algorithmic curation mechanisms that promote emotionally engaging content [30]. This matters for the present study because, on platforms such as Facebook, emotional responses are expressed directly through reaction mechanisms, which provide observable signals of collective audience behavior.

These three lines of research describe how ideological boundaries shape exposure and diffusion, and how cross-cutting encounters can be accompanied by negative affect. They say less, however, about a single sharing event, in which a page shares a link that departs from its ideological sharing repertoire. Two things can be observed at that level. Reactions show how positively a link is received across posts and how homogeneous the response to a particular post is, and the toxicity of the accompanying text shows the language the page itself uses to frame the link. We do not treat reaction homogeneity or toxicity as direct measures of political polarization.

Problem Statement

We consider a set of political news links, each with a content bias score $b_c \in [-1, +1]$ ranging from left- to right-leaning. A link may be shared by one or more Facebook pages, and each page p has a page bias b_p computed from the bias scores of the links it shared, as described in the Political bias metrics subsection (Eq. 2).

Note that b_p is computed from the set of distinct bias-scored news links in the page's sharing history, so repeated shares of the same link add no weight. In this sense, b_p summarizes the page's ideological sharing repertoire. It characterizes the page's observed sharing behavior rather than an independently established or time-invariant political ideology.

For each Facebook post i , in which page p shares news link c , we define Bias Discrepancy $\Delta_{b,i}$ as the absolute difference between b_c and b_p :

$$\Delta_{b,i} = |b_c - b_p|. \quad (1)$$

Larger values thus indicate that the shared news link lies farther from the page's ideological sharing repertoire. Each post also carries the text the page wrote alongside the link, from which we compute toxicity, and receives positive and negative Facebook reactions.

Methodology

This study followed a six-stage analytical process: (1) data collection from Facebook via the CrowdTangle API, (2) data preprocessing, (3) computation of political bias metrics, (4) characterization of post text and shared news through toxicity scoring and news topic classification, (5) computation of audience reaction metrics, and (6) statistical modeling using a Generalized Linear Mixed Model (GLMM) with a Beta distribution. Fig 1 provides an overview of the analytical framework, illustrating the relationships between Facebook pages, the news links they share, and the posts that carry them, together with the Twitter-derived score used in the analysis. It shows how a page's

estimated bias (b_p) and the bias of the shared news link (b_c) combine to produce Bias Discrepancy (Δ_b), and it locates the other measures used in the analysis at the level of the content or individual post, including news topic, Content Reaction Positivity (CRP), toxicity, and the Reaction Homogeneity Index (RHI).

The framework combined information from two platforms with distinct roles. Twitter retweet data provided the ideological signal used to estimate the political bias of each news link (b_c). These same news links were then identified in public Facebook posts, which constituted the observations analyzed in this study. On Facebook, pages' sharing histories were used to estimate page bias (b_p), while individual posts provided the accompanying text and audience reactions analyzed below.

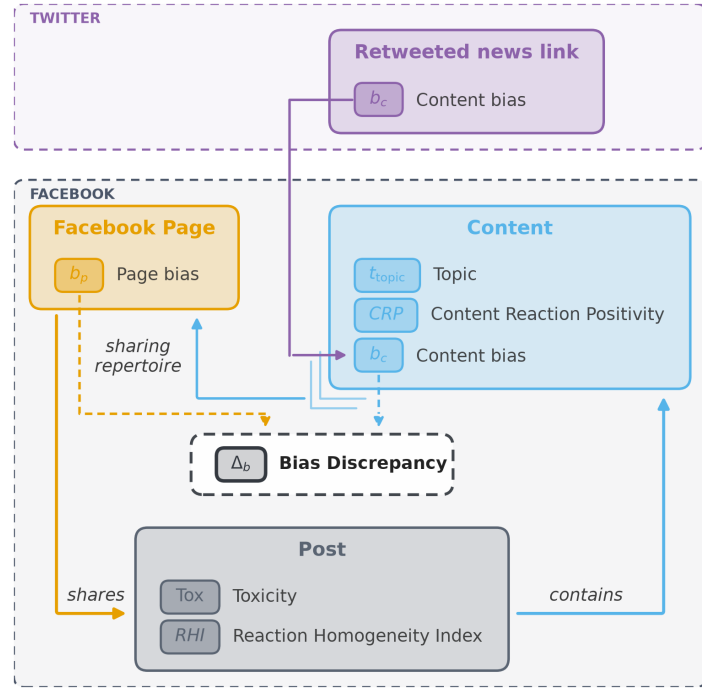


Fig 1. Conceptual framework of the study. The dashed bands distinguish the Twitter and Facebook components of the analysis. On Twitter, each retweeted news link receives a content bias score (b_c) estimated from retweet behavior. The same score is attached to the corresponding content shared on Facebook and is the only measure transferred between platforms. On Facebook, each page is characterized by its page bias (b_p), which summarizes its ideological sharing repertoire and is estimated from the news links it shared. Individual posts connect pages to shared content and are characterized by toxicity and the Reaction Homogeneity Index (RHI). Content-level characteristics include b_c , topic, and Content Reaction Positivity (CRP), with CRP calculated from reactions to other posts sharing the same link. Bias Discrepancy is defined for each Facebook post as the absolute distance between content bias and page bias, $\Delta_b = |b_c - b_p|$. Definitions and ranges of all measures are provided in the Methodology.

Data collection

Twitter data

We used a set of 2,317 unique political news links (URLs) from the dataset associated with [26]. These links originated from Twitter data collected around the 2018 Brazilian

presidential election using election-related hashtags, covering the period from October 1 to December 11, 2018.

Each link was associated with a precomputed political bias score (originally denoted $RP(H)_{\text{content}}$ in [26], hereafter content bias, b_c), ranging continuously from -1.0 (left) to $+1.0$ (right). Values between -0.3 and $+0.3$ were considered center-leaning. These scores were derived from the political polarity of Twitter users who retweeted each news link, rather than from direct ideological annotation of the news articles. The scores were provided with the original dataset and were not independently estimated in this study. A summary of their estimation and validation procedures is provided in S7 Appendix, with further details available in [26].

Facebook data

Using the CrowdTangle API [31], we collected public Facebook posts sharing the selected news links. The collected information included post metadata, textual content, and reaction counts, including Like, Love, Angry, and Sad reactions.

Unlike the Twitter dataset, the Facebook collection did not impose a separate publication-date window. Instead, we searched for public Facebook posts sharing the selected links, regardless of their publication dates.

Connecting Twitter and Facebook data

The two datasets were connected through the URLs of the shared news links. Each Facebook post was associated with the precomputed content bias score (b_c) of the corresponding link. Only this score crossed platforms, and no Twitter user-level information entered the Facebook analysis.

We used these scores as proxies for the ideological orientation of the news links shared on Facebook. This interpretation assumes that the ideological signal captured through Twitter retweet behavior remains informative across platforms, while recognizing that user populations and interaction dynamics may differ between Twitter and Facebook.

Data preprocessing

Four filtering steps were applied in sequence to the 28,181 posts in the initial collection, as summarized in Table 1. First, duplicate entries were removed. Second, we excluded observations lacking information required for the analytical measures. This included posts with inaccessible article text, missing content bias scores, no positive or negative reactions among the four reaction types considered (Like, Love, Sad, and Angry), or an undefined leave-one-out Content Reaction Positivity measure. The latter occurred when the shared link had no other post with at least one positive- or negative-valence reaction. Third, we excluded posts associated with five domains or live-coverage URLs removed during preprocessing. Finally, posts with fewer than 50 total Facebook reactions were excluded. The minimum-reaction cutoff was determined through exploratory analysis of sample-size stability and metric dispersion across increasing cutoff values. An empirical assessment of this threshold selection is provided in S1 Appendix.

Analytical sample

After all filtering steps, the final analytical sample comprised 4,584 Facebook posts from 1,243 distinct pages sharing 761 distinct news links. Their publication dates ranged from January 8, 2018 to November 3, 2022. Most posts were concentrated around the 2018 election: 4,341 (94.7%) were published in 2018, and 4,099 (89.4%) fell within the

Table 1. Data cleaning pipeline and post retention at each step.

Step	Filtering procedure	Remaining (N)	Loss (%)
–	Raw posts (initial collection)	28,181	–
1	Duplicates removed	25,473	9.61%
2	Posts with incomplete analytical information removed	17,973	29.44%
3	Posts from excluded domains removed	11,424	36.44%
4	Posts with fewer than 50 reactions excluded	4,584	59.87%

Twitter collection window. Of these, 368 were published between October 1 and the first-round vote, 3,031 between the two rounds, and 700 between the second-round vote and December 11.

Of the remaining posts, 222 (4.8%) preceded the Twitter collection window, 20 (0.4%) followed it within 2018, and 243 (5.3%) were published from 2019 to 2022. Although the Facebook data were not restricted to that window, the large majority of the analytical sample fell within it. Posts outside this period still shared links whose b_c scores were estimated from the 2018 Twitter data.

A page’s sharing history is the set of distinct links it shared in the deduplicated collection, before the filtering steps described above. Its bias-scored links are those with a b_c score, and b_p was computed from them. Sharing histories varied in size: the number of distinct links per page ranged from 1 to 288, with a median of 3 and a mean of 7.3. Of the 1,243 pages, 364 (29.3%) had only one distinct link in their sharing history and contributed 371 posts (8.1%) to the analytical sample.

The interval over which a page was represented in the analytical sample also varied with the number of posts observed. For 665 pages (53.5%), the sample contained a single post and therefore defined no observation interval. Among the 578 pages with more than one post, the median interval between the first and last observed posts was 21 days (interquartile range: 10.6–61.5 days), and 63.7% spanned 30 days or less. That is, for most pages observed more than once, the analytical sample covered weeks, not years. Publication dates, on the other hand, were not available for the full sharing history.

Political bias metrics

To operationalize ideological discrepancy between pages and shared news links, we defined two metrics: page bias (b_p) and Bias Discrepancy (Δ_b).

We estimated each page’s bias (b_p) by extending the polarization heuristic introduced by [26] from hashtag-based to content-sharing-based estimation. Specifically, b_p summarizes the composition of a page’s ideological sharing repertoire based on the bias scores (b_c) of the news links it shared. Prior work evaluated the original, hashtag-based heuristic in two political contexts, Brazil and Canada, reporting substantial agreement among the independent annotators who labeled the seed hashtags and retweet networks that are assortative by estimated political orientation [26]. The link-based extension, however, is new to this study, and we assessed its face validity as described below.

Here, “content” refers to the shared news link, whose content bias score was derived from Twitter retweet networks. It does not include the text or commentary added by the page alongside the link. The page bias score was computed as follows:

$$b_p = \frac{n_R w_R - n_L w_L}{n_L w_L + n_C w_C + n_R w_R}, \quad (2)$$

where n_L , n_C , and n_R denote the numbers of distinct bias-scored news links classified as left-, center-, and right-leaning, respectively, in the page’s sharing history. The page-specific weights w_R , w_L , and w_C are defined as $w_R = \frac{n_R + n_C}{N}$, $w_L = \frac{n_L + n_C}{N}$, and $w_C = \frac{n_R + n_L}{N}$, where $N = n_R + n_L + n_C$ denotes the total number of bias-scored

links. When all of them were classified as center-leaning ($n_L = n_R = 0$), the denominator of Eq. 2 is zero. We therefore defined $b_p = 0$ for this degenerate case. Values of b_p approaching +1.0 indicate an ideological sharing repertoire dominated by right-leaning content, whereas values approaching -1.0 indicate a repertoire dominated by left-leaning content.

The informativeness of b_p as a measure of a page’s ideological sharing repertoire depends on the number of distinct links observed for that page. In the limiting case of a page represented by only one distinct link, b_p is determined entirely by the ideological category of that link and can therefore take only -1, 0, or +1 under our weighting scheme. In such cases, b_p should not be interpreted as a stable estimate of the page’s broader ideological orientation, but rather as a characterization of its observed sharing behavior within the available data. We assessed this limitation through a sensitivity analysis that excluded pages represented by only one distinct link, reported in S5 Appendix.

To assess the face validity of b_p , we examined all 117 pages in the analytical sample with at least 15 distinct shared links. We assigned Left, Center, or Right labels from the identity of named political actors or movements, explicit ideological self-descriptions, and the editorial identity of recognizable media outlets, without consulting b_p . This yielded labels for 66 pages, while the other 51 could not be classified from their identities. We then categorized b_p using cutoffs at -0.3 and +0.3. The identity-based and b_p -based categories matched for 62 of the 66 classified pages (93.9%), and the continuous b_p scores tracked the ordered identity-based labels (Spearman’s $\rho = 0.897$). Agreement was lower for media outlets (15/18), whose labels involved editorial judgment. The coding rules and category-level results are reported in S4 Appendix.

With b_p operationalized, we computed the Bias Discrepancy (Δ_b), defined in Eq. 1, to quantify how far a particular shared link lies from a page’s ideological sharing repertoire. Values near 0 indicate that the link is similar in ideological orientation to that repertoire, while larger values indicate a greater departure from it. Because b_p was estimated from sharing history and not from an independent assessment of the page’s ideology, Δ_b measures how far a share falls from that repertoire, not its incongruence with an established page position. It cannot distinguish, on its own, whether such a share reflects surprise, criticism, attention seeking, or another motive. We computed one Δ_b value for each Facebook post and examined its association with audience reactions and toxicity.

Toxicity and topic characterization

We also characterized each Facebook post with two measures taken from different texts. First, we analyzed the linguistic tone of the text written by the page alongside the shared link using toxicity scoring. Second, we scored the linked news article on ten topics using zero-shot classification.

Toxicity (Tox) captures the presence of hostile or inflammatory language in the text written by the page alongside the shared news link. To measure it, we employed Google’s Perspective API [32], which returns a continuous score $\text{Tox} \in [0, 1]$ indicating the likelihood that a text may be perceived as toxic (e.g., insults, threats, or hate speech). We processed the Facebook post text directly in its original language (Portuguese), rather than translating it into English. This decision followed findings by [33], who showed that automatic translation from Brazilian Portuguese to English can artificially reduce toxicity scores, particularly for highly toxic comments.

To characterize the thematic content of the shared news articles, we employed the multilingual transformer model mDeBERTa-v3-base for zero-shot topic classification. The procedure used to define the topic categories and select the final classification model is described in S2 Appendix. The resulting topic scores, one per topic and article,

allowed us to account for differences in the subject of the news when modeling reaction homogeneity.

Audience reaction metrics

We next defined two metrics derived from Facebook reaction counts. They describe different aspects of audience response: Content Reaction Positivity (*CRP*) characterizes how a shared news link is received across other posts carrying the same link, whereas the Reaction Homogeneity Index (*RHI*) characterizes the balance of positive and negative reactions on the focal post.

Reaction counts on Facebook provide observable signals of audience responses to political content [29]. We classified Like and Love as positive-valence reactions and Sad and Angry as negative-valence reactions. Haha and Wow were excluded because their valence is more context dependent and cannot be mapped consistently onto a positive–negative axis. The four reaction types retained in the analysis accounted for 82.09% of all reactions in the analytical sample. To characterize how the shared news link was received in other posts, we defined Content Reaction Positivity (*CRP*) from reactions to all other posts sharing link c , excluding the focal post i . Let $\mathcal{P}_c$ denote the set of posts sharing c . Then:

$$CRP_i = \frac{\sum_{\substack{j \in \mathcal{P}_c \\ j \neq i}} n_j^+}{\sum_{\substack{j \in \mathcal{P}_c \\ j \neq i}} (n_j^+ + n_j^-)},$$

where n_j^+ and n_j^- denote the counts of positive reactions (Like, Love) and negative reactions (Angry, Sad) for post j . The score ranges from 0 to 1, with higher values indicating predominantly positive reactions to the shared link in other posts. CRP_i describes reactions to the same link elsewhere in the dataset, not reactions to the focal post.

For a focal post to have a defined leave-one-out Content Reaction Positivity score, the same news link must be represented by at least one other post with at least one positive- or negative-valence reaction. Observations for which no such comparison post was available were excluded during preprocessing.

To characterize the balance of positive and negative reactions on the focal post, we defined the Reaction Homogeneity Index (*RHI*):

$$RHI_i = \frac{|n_i^+ - n_i^-|}{n_i^+ + n_i^-}.$$

Values close to 1 indicate that reactions are predominantly of one sign, whether positive or negative, whereas values near 0 indicate that positive and negative reactions occur in similar numbers. The index characterizes the distribution of reaction types and does not measure agreement among users, their political positions, or political polarization.

Both measures were assigned to each focal post but were computed from different sets of reactions. CRP_i summarizes reactions to other posts sharing the same news link ($j \neq i$), whereas RHI_i uses only reactions to the focal post i . A post with predominantly negative reactions can therefore have a high RHI_i , as can a post with predominantly positive reactions, and a roughly even mix produces a low RHI_i . If $f_i^- = n_i^- / (n_i^+ + n_i^-)$ denotes the focal post’s negative-reaction share, then $RHI_i = |1 - 2f_i^-|$. In this sample, 4,347 posts (94.8%) received at least as many positive as negative reactions ($f_i^- \leq 0.5$),

so $RHI_i = 1 - 2f_i^-$ for these observations. In other words, for the large majority of posts, lower RHI is mathematically a larger negative-reaction share. The negative-share analyses in S5 Appendix are therefore another view of the same reaction-composition pattern, not independent corroborating evidence. By contrast, no such identity holds between CRP_i and RHI_i , which are computed from different posts. Their observed correlation ($r \approx 0.75$) is therefore an empirical association and not evidence that reaction positivity causes reaction homogeneity.

Statistical modeling

Main model

To address RQ2, we estimated the association between Bias Discrepancy and reaction homogeneity while accounting for other page, content, and post characteristics. Our main model was a Generalized Linear Mixed Model (GLMM) with a Beta distribution and logit link.

The fixed effects included signed content and page bias (b_c and b_p), their absolute values ($|b_c|$ and $|b_p|$), Bias Discrepancy (Δ_b), toxicity, Content Reaction Positivity (CRP), ten topic scores, and controls for total reactions and page links. Signed bias captures left–right direction, whereas absolute bias captures distance from the ideological center. Including these terms alongside Δ_b allowed us to estimate the discrepancy association while accounting for the direction and extremity of page and content bias. This representation also provided the best fit among the bias representations considered and was retained in the main model.

The topic scores were included as separate continuous predictors because a news article may be associated with multiple topics. The activity controls were log-transformed: total reactions included all reaction types recorded for the focal post, and page links counted distinct links in the page’s sharing history. By contrast, the valence-based reaction measures used only Like, Love, Sad, and Angry. Finally, including CRP accounted for reactions to the same link in other posts. Its coefficient is read as an empirical association, not as an independent behavioral effect.

The data had a cross-classified structure: pages could contribute multiple posts, and the same news link could be shared by multiple pages. We therefore included random intercepts for pages and content (shared news links). Random slopes for toxicity at both levels allowed its association with reaction homogeneity to vary across pages and links.

Because the Beta distribution is defined on $(0, 1)$, we transformed the observed RHI values, including those at 0 and 1, into the open interval using the adjustment proposed by Smithson and Verkuilen [34]. The transformed outcome is denoted RHI' , with details provided in S3 Appendix.

The main model was estimated using `glmmTMB` in R:

$$\begin{aligned} RHI' \sim & b_c + |b_c| + b_p + |b_p| + \Delta_b + \text{Tox} + CRP \\ & + \log(\text{total reactions}) + \log(\text{page links}) \\ & + t_{\text{Economy}} + t_{\text{Education}} + t_{\text{Health}} + t_{\text{Security}} \\ & + t_{\text{Culture}} + t_{\text{Religion}} + t_{\text{Disinformation}} \\ & + t_{\text{Election}} + t_{\text{Politics}} + t_{\text{Corruption}} \\ & + (\text{Tox} \mid \text{Page}) + (\text{Tox} \mid \text{Content}). \end{aligned}$$

Here, each $t_{\text{topic}} \in [0, 1]$ is the zero-shot classifier score assigned to that topic for the shared news link.

Estimation and diagnostics

All continuous predictors were standardized using Z-score normalization within each fitted sample, and the boundary adjustment of RHI was recomputed using that sample's size. Because the continuous predictors were standardized, we compared their associations with RHI by the absolute size of their coefficients. Coefficient confidence intervals were two-sided 95% Wald intervals. Marginal and conditional R^2 were calculated using the Nakagawa method implemented in `performance::r2_nakagawa()`, with the lognormal approximation.

Because Δ_b is derived from b_p and b_c , we assessed potential multicollinearity using `performance::check_collinearity()`. All VIFs were below 5 in the main model, and the maximum across the sensitivity models was 5.10.

Sensitivity analyses

We compared three bias representations: signed content and page bias alone, signed bias together with absolute extremity, and Left/Center/Right categories. These comparisons used the same posts and remaining covariates but omitted the two activity controls.

We also refit the main model after omitting CRP , restricting pages to at least two or five distinct bias-scored links, restricting posts to the Twitter collection window, recalculating page bias without any shares of the focal link by the page, and replacing the minimum-reaction cutoff of 50 with 25 or 100.

Finally, we analyzed the focal post's negative-reaction share, $f_i^- = n_i^- / (n_i^+ + n_i^-)$, using a mixed beta model and a beta-binomial model of the underlying reaction counts. Both retained the main model's other terms while omitting CRP . Full model-comparison and sensitivity details are reported in S5 Appendix.

Results

We first describe the sample and the distributions of the analytical variables. Because Bias Discrepancy combines page bias and content bias, we then examine how each of the two relates to audience reactions and toxicity, before addressing RQ1 through the analysis of Bias Discrepancy and RQ2 through the main model.

Sample composition and variable distributions

Table 2 shows that both shared news links and pages were skewed toward the left. Most posts shared left-leaning links (52.79%), followed by center-leaning (31.48%) and right-leaning links (15.73%). Pages were also predominantly left-leaning, although the ordering differed: left-leaning pages accounted for 54.54% of the total, right-leaning pages represented 25.90%, and center-leaning pages were the least frequent group (19.55%).

Fig 2 shows the distributions of the main analytical variables. Content Reaction Positivity (CRP) was concentrated near 1.0, indicating that reactions to shared links in other posts were predominantly positive (Like, Love). Bias Discrepancy (Δ_b) was concentrated at lower values, indicating that most posts shared content relatively close to the page's ideological sharing repertoire. Toxicity (Tox) had a right-skewed distribution, with most posts having low toxicity scores and a smaller number extending toward higher values. Finally, Reaction Homogeneity Index (RHI) values were concentrated near 1.0, indicating that reactions on most posts were predominantly of one sign rather than evenly divided between positive and negative reactions.

Fig 3 shows the distribution of topic scores across the shared news links. Politics and elections (t_{Politics} , t_{Election}) had the highest median scores and wide variability,

Table 2. Distribution of content bias categories across Facebook posts and page bias categories across distinct pages.

Category	N	Percentage
<i>Content bias (N = 4,584 posts)</i>		
Left ($b_c \leq -0.3$)	2,420	52.79%
Center ($-0.3 < b_c < 0.3$)	1,443	31.48%
Right ($b_c \geq 0.3$)	721	15.73%
<i>Page bias (N = 1,243 pages)</i>		
Left ($b_p \leq -0.3$)	678	54.54%
Center ($-0.3 < b_p < 0.3$)	243	19.55%
Right ($b_p \geq 0.3$)	322	25.90%

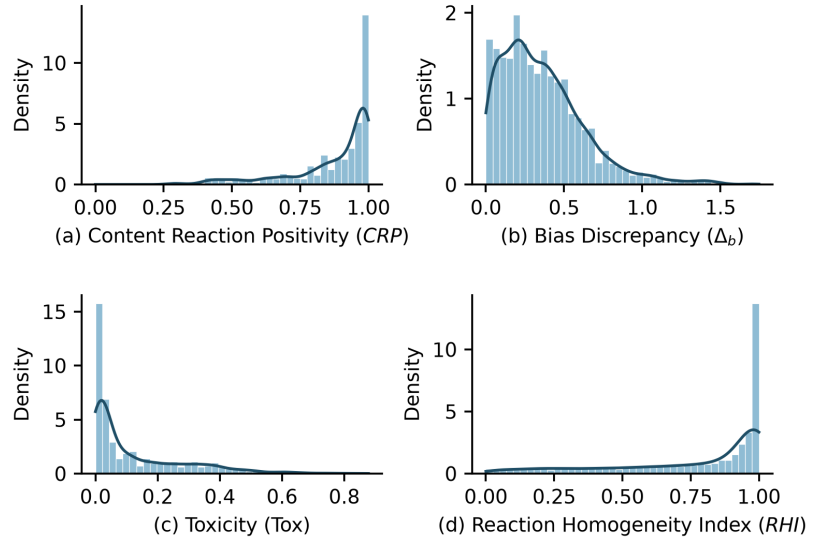


Fig 2. Distributions of the main analytical variables. Curves represent kernel density estimates.

indicating that these topics were common across the corpus. In comparison, education and religion ($t_{\text{Education}}$, t_{Religion}) had lower median scores, although high-score observations indicated that they were prominent in some news articles.

Association between political bias and audience reactions

Fig 4 illustrates how page bias (b_p) and content bias (b_c) were associated with Content Reaction Positivity, the Reaction Homogeneity Index, and the toxicity of the accompanying post text. Statistical significance was assessed using Welch’s ANOVA and Games–Howell post-hoc tests to account for unequal variances.

Content Reaction Positivity (Fig 4, panel (a)): Content Reaction Positivity varied systematically across ideological groups. Links shared by left-leaning pages had lower average Content Reaction Positivity than those shared by center- and right-leaning pages ($p < 0.001$). Across content categories, right-leaning links had higher average Content Reaction Positivity than left- and center-leaning links ($p < 0.05$).

Reaction Homogeneity Index (Fig 4, panel (b)): Reaction homogeneity varied across combinations of page and content bias. For center-leaning content, posts by left-leaning pages had lower *RHI* than posts by center- and right-leaning pages (both $p < 0.001$). For right-leaning content, *RHI* increased from left- to center- to right-leaning pages,

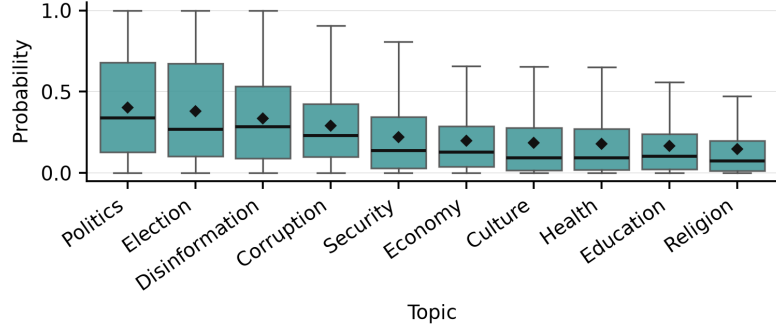


Fig 3. Distribution of zero-shot classification scores across topics. Boxplots represent the score assigned to each topic across shared news links.

with all pairwise differences significant ($p < 0.001$). Within left-leaning pages, however, *RHI* did not differ significantly across content-bias categories ($p = 0.286$).

The lowest observed mean *RHI* occurred when left-leaning pages shared right-leaning content, although, as noted above, differences across content-bias categories within left-leaning pages were not statistically significant. Posts by right-leaning pages sharing left-leaning content also had lower mean *RHI* than those sharing right-leaning content. These comparisons describe the distribution of reaction types and do not establish whether individual users encountered opposing views or disagreed about the political meaning of the content.

Toxicity (Fig 4, panel (c)): Toxicity patterns differed from those observed for Content Reaction Positivity and the Reaction Homogeneity Index. Center-leaning pages had lower toxicity than both left- and right-leaning pages across content categories (all $p < 0.01$). With respect to content bias, we found no consistent evidence that posts sharing ideologically opposing content contained more toxic text than posts sharing aligned content, despite variation in median toxicity across groups.

Page and content bias were therefore associated with differences in Content Reaction Positivity and reaction homogeneity. Among these differences, posts by right-leaning pages generally had more homogeneous reactions in this sample. This pattern is compatible with prior work reporting stronger cohesion within right-leaning communities in the Brazilian context [11,26], although reaction homogeneity in our data does not itself measure audience cohesion or political agreement.

Association between Bias Discrepancy, audience reactions, and toxicity

The comparisons above used bias categories. Bias Discrepancy instead measures, on a continuous scale, how far a shared link lies from the page’s ideological sharing repertoire. To examine how it related to audience reactions and toxicity (RQ1), we compared Content Reaction Positivity (*CRP*), the Reaction Homogeneity Index (*RHI*), and Toxicity (*Tox*) across quartiles of Bias Discrepancy (Δ_b), from “Very Low” (strong alignment with that repertoire) to “Very High” (strong departure from it). Statistical significance was assessed using Welch’s ANOVA and Games–Howell post-hoc tests.

The quartile cut points for Δ_b were 0.161, 0.313, and 0.498. The four groups contained 1,148, 1,148, 1,142, and 1,146 posts, respectively, and their similar sizes follow from the quartile definition. Each group included hundreds of distinct pages and news links, indicating that the observed discrepancy patterns were not confined to a small number of pages or news links. The Very High discrepancy group, for instance, included

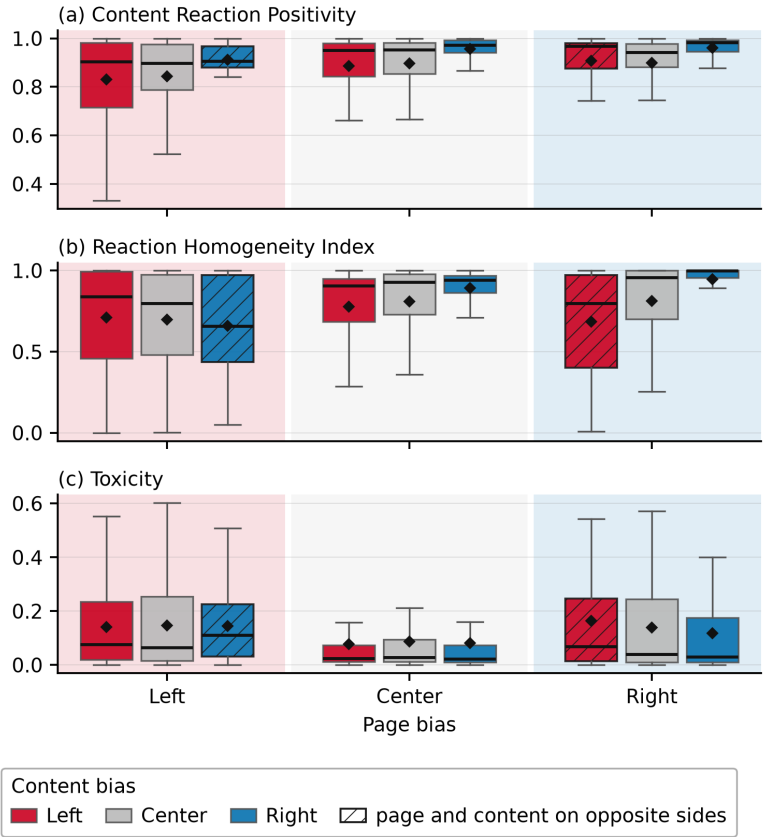


Fig 4. Distribution of Content Reaction Positivity, Reaction Homogeneity Index, and Toxicity across page and content bias categories. Panels (a) to (c) show Content Reaction Positivity, the Reaction Homogeneity Index, and Toxicity, respectively. The background band of each group indicates page bias and the box fill indicates content bias. Boxes whose fill contrasts with the background band represent posts in which page and content fall on opposite ideological sides and are shown with hatching.

595 distinct pages, with no single page accounting for more than 3.9% of posts in that group. Page concentration by group is reported in S6 Appendix.

Content Reaction Positivity (Fig 5, panel (a)): Content Reaction Positivity was significantly lower in the “Very Low” Bias Discrepancy category than in all other levels of Bias Discrepancy ($p < 0.001$). Put differently, links closely aligned with a page’s ideological sharing repertoire tended to receive a lower proportion of positive reactions in other posts sharing the same link. The data do not show what produced this unexpected pattern, and we interpret it with caution.

Reaction Homogeneity Index (Fig 5, panel (b)): Reaction homogeneity was significantly lower in the “Very Low” and “Very High” Bias Discrepancy categories than in the “Low” and “High” categories. No statistically significant difference was observed between the “Very Low” and “Very High” groups. The relationship between Bias Discrepancy and reaction homogeneity was therefore non-monotonic, with lower homogeneity at both ends of the discrepancy distribution.

Toxicity (Fig 5, panel (c)): Toxicity also differed across Bias Discrepancy levels ($p < 0.001$). No statistically significant differences were observed among the lower and moderate discrepancy groups. The “Very High” category, however, had significantly higher toxicity than all other groups. This does not conflict with the category

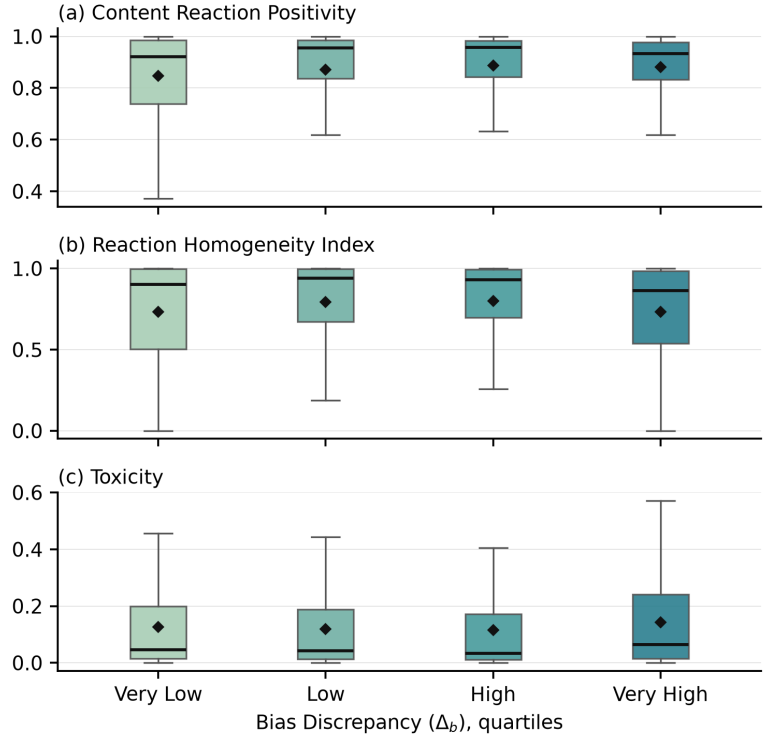


Fig 5. Distribution of Content Reaction Positivity, Reaction Homogeneity Index, and Toxicity across quartiles of Bias Discrepancy (Δ_b), from “Very Low” to “Very High.” Panels (a) to (c) show Content Reaction Positivity, the Reaction Homogeneity Index, and Toxicity, respectively.

comparison above, which found no consistent toxicity difference between posts sharing opposite-side and aligned content. Every post pairing a page and a link on opposite sides fell in the Very High group, but these were only 176 of its 1,146 posts. Most of the group paired a center-leaning page with a left- or right-leaning link, or the reverse (733 posts), and the rest were large departures within the same side (237 posts).

The three outcomes thus related to Bias Discrepancy in different ways. For RQ1, the greatest departures from a page’s ideological sharing repertoire were associated with less homogeneous reactions and more toxic accompanying text, whereas the lowest Content Reaction Positivity occurred at the opposite end of the discrepancy distribution.

Modeling factors associated with reaction homogeneity

The quartile comparisons in the preceding section describe the unadjusted relationship between Bias Discrepancy and reaction homogeneity, with lower *RHI* at both ends of the observed discrepancy distribution. The mixed model below answers a different question. It estimates the association between continuous Δ_b and *RHI* conditional on the remaining covariates and on clustering by Facebook page and shared news link. Because Δ_b enters the model as a linear term on the logit scale, its coefficient summarizes an adjusted association across the observed discrepancy range. It does not imply that the unadjusted relationship is monotonic.

Table 3 presents the main model predicting the Reaction Homogeneity Index, which separates signed bias from absolute extremity and controls for total reactions and page links. Its marginal and conditional R^2 were 0.762 and 0.977, respectively.

Table 3. Main model: mixed beta regression predicting the Reaction Homogeneity Index. Continuous predictors are standardized. The model includes page and content random intercepts and random slopes for toxicity.

Predictor	β	95% CI	p
Intercept	1.536	1.464 – 1.607	< 0.001
Content bias (b_c)	0.158	0.081 – 0.236	< 0.001
Content extremity ($ b_c $)	0.072	0.002 – 0.143	0.044
Page bias (b_p)	0.099	0.043 – 0.155	< 0.001
Page extremity ($ b_p $)	0.114	0.059 – 0.168	< 0.001
Bias Discrepancy (Δ_b)	-0.173	-0.211 – -0.135	< 0.001
Toxicity (Tox)	-0.149	-0.196 – -0.102	< 0.001
Content Reaction Positivity (CRP)	0.951	0.888 – 1.013	< 0.001
log(total reactions)	-0.115	-0.159 – -0.071	< 0.001
log(page links)	-0.071	-0.126 – -0.017	0.010
t_{Economy}	0.034	-0.045 – 0.114	0.397
$t_{\text{Education}}$	0.113	0.032 – 0.193	0.006
t_{Health}	-0.096	-0.190 – -0.003	0.044
t_{Security}	0.079	-0.023 – 0.181	0.128
t_{Culture}	-0.093	-0.203 – 0.017	0.098
t_{Religion}	-0.018	-0.099 – 0.063	0.658
$t_{\text{Disinformation}}$	-0.101	-0.195 – -0.006	0.036
t_{Election}	-0.011	-0.110 – 0.088	0.831
t_{Politics}	0.085	-0.026 – 0.197	0.134
$t_{\text{Corruption}}$	0.017	-0.087 – 0.122	0.743
Posts / pages / links	4,584 / 1,243 / 761		
Marginal / conditional R^2	0.762 / 0.977		
AIC	-15046.2		

Greater Bias Discrepancy was associated with lower reaction homogeneity ($\beta = -0.173$, $p < 0.001$), as was toxicity ($\beta = -0.149$, $p < 0.001$). By contrast, Content Reaction Positivity (CRP), which summarizes reactions to the same link in other posts, was positively associated with RHI ($\beta = 0.951$). Both activity controls had negative coefficients: posts with more reactions and posts from pages sharing more distinct links tended to have lower RHI , conditional on the other model terms.

For RQ2, Content Reaction Positivity therefore had the largest standardized coefficient. Bias Discrepancy had the next largest, followed by content bias ($\beta = 0.158$) and toxicity. The confidence intervals of these three overlapped, so their relative order was not well determined.

Across the main model and seven sensitivity analyses, the standardized Bias Discrepancy coefficient remained negative, ranging from -0.219 to -0.167 , with all 95% confidence intervals excluding zero. The association persisted when CRP was omitted ($\beta = -0.178$, 95% CI $[-0.219, -0.138]$) and when page bias was recalculated without the focal shared link ($\beta = -0.192$, 95% CI $[-0.234, -0.150]$). It also persisted under the restrictions on bias-scored links, restriction to the Twitter collection window, and alternative minimum-reaction cutoffs. Toxicity likewise retained a negative coefficient in all specifications, with all 95% confidence intervals excluding zero. Full results are reported in S5 Appendix.

To provide concrete illustrations of the relationship between Bias Discrepancy and reaction homogeneity, we examined high-discrepancy observations from pages represented by at least 15 distinct links and selected examples in both ideological directions. For *A Gazeta Esquerdista Contra o Coiso*, whose ideological sharing repertoire was strongly left-leaning ($b_p = -0.833$), sharing a right-leaning link ($b_c = +0.615$) produced a large Bias Discrepancy ($\Delta_b = 1.449$). The post had an RHI of 0.135, below the page’s mean RHI of 0.739.

Conversely, *Grupo do Movimento Brasil à Direita* had a right-leaning ideological sharing repertoire ($b_p = +0.580$) but shared a left-leaning link ($b_c = -0.749$), resulting in a high Bias Discrepancy ($\Delta_b = 1.329$). The page shared this same link in two separate posts, which drew different reactions. The post with higher toxicity

(Tox = 0.47) had a relatively low *RHI* of 0.375, compared with the page-level mean of 0.842 (median = 1.000). The other post, with a toxicity of 0.01, had an *RHI* of 0.944.

These two examples show that high Bias Discrepancy can coincide with relatively heterogeneous reactions in either ideological direction. The two posts from *Grupo do Movimento Brasil à Direita* also allow a within-page comparison. With the page and the link held fixed, the post with the more toxic accompanying text drew less homogeneous reactions.

A counter-example illustrates that this pattern does not apply to every high-discrepancy post. *Brasil Conservador* ($b_p = +0.681$) shared a strongly left-leaning link ($b_c = -0.749$), producing a similarly large Bias Discrepancy ($\Delta_b = 1.430$), yet the post had highly homogeneous reactions ($RHI = 0.960$), close to the page’s mean *RHI* of 0.951. The negative association between Bias Discrepancy and *RHI* should therefore be interpreted as an aggregate tendency rather than a post-level rule.

Beyond the fixed effects, the random effects describe how pages differed from one another. Fig 6, panel (a), shows the pages with the largest deviations in baseline reaction homogeneity (random intercepts). Pages with strongly negative deviations included both news-oriented pages (e.g., “BBC News Brasil”) and politically oriented pages, indicating that lower baseline reaction homogeneity was not exclusive to a single type of page. The presence of a major news outlet among these cases is consistent with prior work describing some news outlets as *bubble reachers* (or brokers), capable of distributing content across different ideological clusters [26]. Such broader reach could be associated with more heterogeneous audience reaction patterns, although the random effects estimated here do not directly measure audience composition or brokerage.

In contrast, several pages with strongly positive deviations were explicitly political or partisan (e.g., “Jair M. Bolsonaro,” representing the page of the far-right candidate for president of Brazil in 2018), indicating more homogeneous audience reactions after accounting for the characteristics included in the model. This contrast is broadly consistent with prior work showing differences in interaction patterns between more neutral brokers and partisan pages in the Brazilian political context [9]. These page-level examples are, however, only illustrative. The random effects capture residual heterogeneity in reaction homogeneity and do not establish audience ideology, political disagreement, audience composition, or the processes that produce these differences.

Fig 6, panel (b), shows a positive correlation between the estimated page random intercepts and random slopes for toxicity ($r = 0.81$). In other words, pages with higher estimated baseline reaction homogeneity tended to have toxicity slopes above the global average, whereas pages with lower estimated baselines tended to have slopes below it. Note that a slope above the global average can still represent a negative association. This pattern describes page-level heterogeneity in the fitted model. Because many pages had few observed posts, individual page estimates should be interpreted cautiously.

Social Identity Theory [35] offers one interpretation of this correlation, in which hostile language works differently for different political audiences. In an online experiment, like-minded incivility led participants to write more uncivil comments themselves, while disagreeable incivility induced anger and aversion [36]. On social media, posts about the political out-group were shared about twice as often as posts about the in-group [37]. If toxic posts on pages with homogeneous reactions reach mostly like-minded users, toxicity there would be less likely to split reactions. Our data do not show who reads these posts or whether their toxic language targets an out-group, so we treat this correlation as variation across pages in the toxicity association, not as evidence of in-group cohesion.

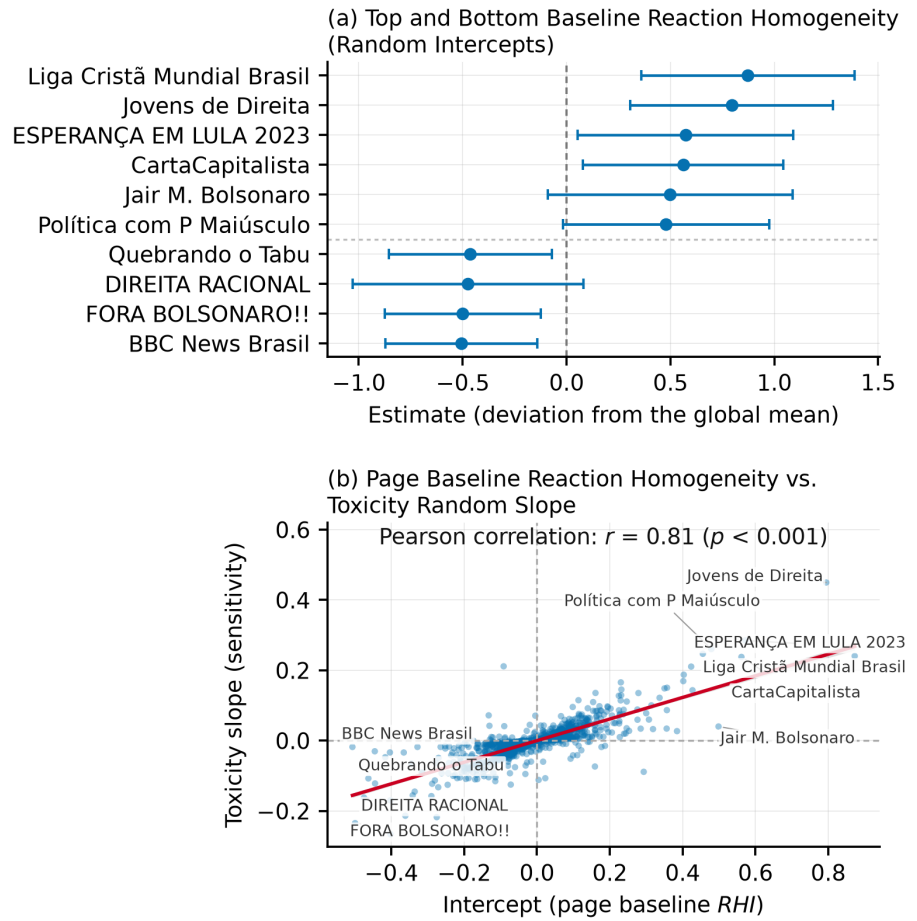


Fig 6. Page-level heterogeneity from the main model. Panel (a) shows pages with the most extreme random-intercept deviations in baseline Reaction Homogeneity Index, with 95% intervals. Panel (b) relates page-level random intercepts to random slopes for toxicity. The two are positively correlated ($r = 0.81$).

Conclusion

This study examined how ideological discrepancy between Facebook pages and shared news links was associated with audience reactions and with the toxicity of the accompanying post text in a dataset centered on Brazil's 2018 presidential election. Cross-cutting content is often expected to mitigate polarization, yet encounters across ideological lines do not necessarily foster more constructive engagement [8,17]. At the level of individual sharing events, our results are consistent with that view. Posts at the upper end of Bias Discrepancy drew less homogeneous reactions than posts in the middle of the distribution and carried the most toxic accompanying text. Departure from the ideological sharing repertoire was not the only condition associated with mixed reactions, however. Reaction homogeneity was also lower when the shared news link was very close to that repertoire, and Content Reaction Positivity was lowest at the same end, although the data do not establish the mechanism underlying this unexpected pattern.

Posts by right-leaning pages also generally had more homogeneous reactions than posts by left-leaning pages. This asymmetry is compatible with prior work reporting stronger cohesion within right-leaning communities in the Brazilian context [11,26], but

reaction homogeneity does not itself measure audience cohesion or political agreement, and the smaller number of right-leaning pages and links limits the generalizability of this comparison.

For RQ2, Content Reaction Positivity, which reflects how the same link was received in other posts, had the largest association with reaction homogeneity, and omitting it reduced the marginal R^2 of the main model from 0.762 to 0.154 (S5 Appendix). Greater Bias Discrepancy remained associated with lower reaction homogeneity after accounting for this link-level reception, page and content bias, ideological extremity, toxicity, content topics, and activity controls. The association between toxicity and reaction homogeneity also varied across pages. Pages with higher baseline reaction homogeneity tended to have more positive toxicity slopes, and pages with lower baseline reaction homogeneity tended to have more negative slopes. Work on social identity and incivility offers a possible explanation [35–37], which our data cannot test.

These results should be interpreted in light of several limitations. First, content bias scores were derived from Twitter retweet behavior [26], whereas Facebook posts and audience reactions were observed on Facebook. This approach assumes that the ideological signal derived from Twitter is informative about the news links when they circulate on Facebook, even though user populations and interaction dynamics may differ across platforms. In addition, Facebook’s reaction affordances may shape both Content Reaction Positivity and the Reaction Homogeneity Index, so their absolute levels should be interpreted in the context of the platform.

Second, page bias is inferred from the ideologically labeled links observed in a page’s sharing history. Bias Discrepancy therefore measures departure from the page’s ideological sharing repertoire, rather than independently established page ideology or the page’s intent in sharing a particular link. Even so, for pages with at least 15 distinct links, the inferred page bias agreed with an independent identity-based classification of page orientation for 62 of the 66 classified pages (see S4 Appendix). In addition, sensitivity analyses restricted pages to at least two or five distinct bias-scored links and recalculated page bias after excluding all shares of the focal link by the page. The negative association between Bias Discrepancy and reaction homogeneity persisted in each analysis (see S5 Appendix). These checks address short observed histories and the focal link’s contribution to the page-bias estimate, but they do not establish independently held page ideology.

Third, right-leaning content and pages are underrepresented in the dataset, requiring caution when generalizing comparisons involving this group. Fourth, two features of the reaction measures constrain their interpretation. Content Reaction Positivity is computed from other posts sharing the same link and captures a content-level reaction pattern that overlaps with the content random intercept. We therefore treat it as a contextual predictor, not as an independent behavioral mechanism. The Reaction Homogeneity Index is mathematically related to the focal post’s negative-reaction share, so results from these two outcomes should not be treated as independent evidence of the same association.

Fifth, the temporal relationship between the Twitter and Facebook data imposes additional limitations. Content bias scores were estimated from Twitter retweet behavior between October 1 and December 11, 2018. Of the Facebook posts in the analytical sample, 4,099 (89.4%) fell within this window, 222 (4.8%) preceded it, 20 (0.4%) followed it later in 2018, and 243 (5.3%) were published from 2019 to 2022. Restricting the model to the 4,099 posts within that window preserved the negative association between Bias Discrepancy and reaction homogeneity (see S5 Appendix). Nevertheless, for posts outside this window, b_c is a link-level ideological proxy estimated from 2018 Twitter circulation rather than a contemporaneous measure of the link’s ideological reception. Because the Facebook data were collected retrospectively, the

sample may also be affected by survivorship if deleted or otherwise inaccessible posts could not be observed. CrowdTangle further limits the analysis to the public Facebook content available through its collection system, excluding private interactions.

Finally, toxicity was measured using the Perspective API, a widely used tool with known limitations for Portuguese-language content. Prior work documents lower classification performance for Portuguese than for English [38], as well as limitations in capturing conversational context, sarcasm, and culturally specific forms of political expression [39,40]. Processing the posts directly in Portuguese avoids translation-related distortions [33,41], but the resulting toxicity scores should nevertheless be interpreted as approximate indicators of hostile language rather than exhaustive measures of political incivility [42].

Future research could extend this work using more context-aware and culturally grounded measures of toxicity and by examining whether similar relationships between ideological discrepancy, audience reactions, and accompanying language emerge in other political contexts and on platforms with different interaction affordances.

Ethical considerations

The Facebook data analyzed in this study were collected from publicly accessible posts using the now-deprecated CrowdTangle API. The Twitter-derived content bias scores were obtained from the previously published dataset of [26]. Data collection was conducted prior to the tool’s discontinuation, in compliance with the platform’s Terms of Service and data use policies active at the time of access. The dataset comprises content exclusively from public pages (e.g., media outlets, political figures, and organizations) operating in the public sphere. While we report actor-specific heterogeneity to describe how distinct editorial lines or political stances relate to reaction homogeneity, this analysis does not expose private user data. No personally identifiable information from private citizens was analyzed individually. Toxicity scores were computed exclusively from the publicly available text published by these pages, while audience-related metrics, such as the Reaction Homogeneity Index, were computed from aggregate reaction counts without identifying individual users.

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Supporting Information

S1 Appendix

Empirical assessment of the minimum-reaction cutoff

This appendix presents the empirical analysis used to determine the minimum-reaction cutoff applied in the preprocessing stage.

As shown in Fig 7, panel (a), increasing the minimum-reaction cutoff reduced the number of retained posts, with a less steep decline at higher cutoffs. Panel (b) shows that the standard deviation of Content Reaction Positivity increased as the cutoff rose toward 50 reactions and changed relatively little at higher cutoffs. These curves describe changes in sample size and between-post dispersion. They do not directly measure the precision of individual scores. The 50-reaction cutoff was an exploratory preprocessing choice. Sensitivity analyses using cutoffs of 25 and 100 are reported in S5 Appendix.

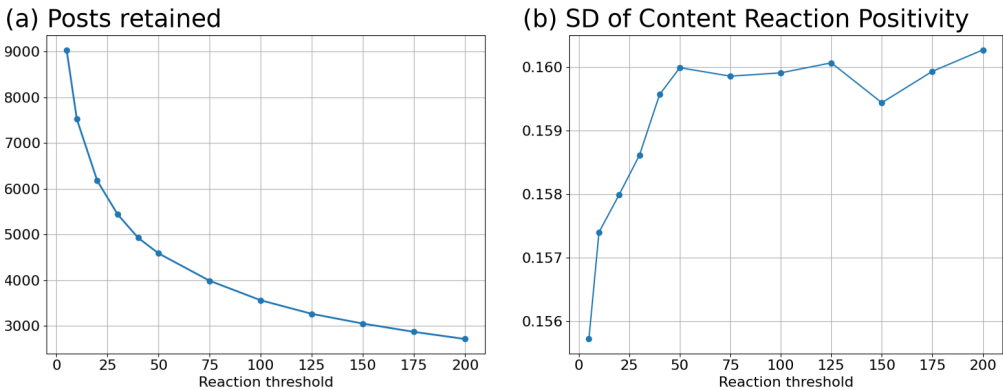


Fig 7. Assessment of the minimum-reaction cutoff. Panel (a) shows the number of retained posts at each cutoff. Panel (b) shows the standard deviation of Content Reaction Positivity across the retained posts. Dispersion rises toward a cutoff of 50 reactions and changes relatively little thereafter.

S2 Appendix

Topic definition and model selection

To characterize the thematic scope of the shared news articles, we first conducted an exploratory Latent Dirichlet Allocation (LDA) analysis [43] on the textual content of the linked news articles. The resulting latent themes informed the definition of ten topical categories used in the analysis: economy, education, health, security, culture, religion, disinformation, election, politics, and corruption. These predefined categories were then assigned to individual news articles using a zero-shot language model.

To select an appropriate zero-shot classification model, we conducted a comparative experiment with three candidate approaches. The first was the multilingual mDeBERTa-v3-base model fine-tuned on NLI data [44]

(MoritzLaurer/mDeBERTa-v3-base-xnli-multilingual-nli-2mil7)¹. The second was the BGE-M3 model [45] (MoritzLaurer/bge-m3-zeroshot-v2.0)². The third followed a translate-test pipeline using BART-Large-MNLI [46] (facebook/bart-large-mnli)³, where texts were first translated from Portuguese to English using the uncamp-dl/translation-pt-en-t5 model [47].

For an exploratory model comparison, we manually labeled 50 news links for the presence of the 10 predefined topics. Of these, 48 matched the analytical news-link inputs used in the comparison, and the reported metrics were calculated on those 48 articles. Each model produced classifier scores $p_k \in [0, 1]$ for topic k . Instead of applying a single global threshold, we optimized the classification threshold independently for each topic to account for differences in model sensitivity across domains. Binary predictions were defined as

$$\hat{y}_k = \begin{cases} 1, & \text{if } p_k \geq \tau_k, \\ 0, & \text{otherwise.} \end{cases}$$

For each topic k , the optimal threshold τ_k was selected by maximizing the F1-score on the validation set. We evaluated the F1-score at every unique probability value produced by the model, which avoids the discretization of a fixed-step grid search.

The thresholds and reported F1 scores were obtained from the same 48 labeled articles, without a separate test set. These scores therefore summarize an exploratory comparison and may overestimate performance on new articles. These thresholds were used only to compare the candidate models. The main model used the continuous scores.

Model performance was summarized using the Weighted Average F1-score across topics. This aggregation accounts for class imbalance by weighting topic-level F1-scores according to their support (number of positive instances) in the validation set.

As summarized in Table 4, the multilingual mDeBERTa-v3-base model achieved the best performance, with a Weighted F1-score of 0.825, compared to 0.780 for the translation-based BART-Large-MNLI pipeline and 0.768 for BGE-M3. Based on this evaluation, we selected mDeBERTa-v3-base for topic classification in the main analysis.

Table 4. Exploratory comparison of zero-shot topic classifiers using 48 matched manually labeled articles. Topic-specific thresholds were selected and F1 scores calculated on the same articles.

Model Strategy	Approach	W-F1 Score
mDeBERTa-v3-base	Multilingual	0.825
BART-Large-MNLI	Trans. (PT→EN)	0.780
BGE-M3	Multilingual	0.768

S3 Appendix

Choice of the beta regression framework

The Reaction Homogeneity Index (*RHI*) is a continuous variable bounded in the unit interval $[0, 1]$. Applying Ordinary Least Squares (OLS) regression to such data is problematic, as it violates key assumptions of homoscedasticity and normally distributed residuals. In bounded outcomes, variance depends on the mean and decreases as observations approach the boundaries of the support.

We therefore used a Beta regression framework within a Generalized Linear Mixed Model (GLMM). The Beta distribution is commonly used for proportions and rates. Its

¹<https://huggingface.co/MoritzLaurer/mDeBERTa-v3-base-xnli-multilingual-nli-2mil7>

²<https://huggingface.co/MoritzLaurer/bge-m3-zeroshot-v2.0>

³<https://huggingface.co/facebook/bart-large-mnli>

conditional variance depends on the mean, and its predicted values stay within the bounds of the outcome.

Boundary adjustment for the Reaction Homogeneity Index

A limitation of the Beta distribution is that it is defined on the open interval (0, 1). However, the empirical Reaction Homogeneity Index included observations with values exactly equal to 0 or 1. Because the likelihood function of the Beta distribution is undefined at these boundaries, a transformation was required prior to estimation.

Following Smithson and Verkuilen [34], we applied a symmetric transformation that compresses the data into the open interval:

$$RHI' = \frac{RHI(N - 1) + 0.5}{N},$$

where *RHI* is the original Reaction Homogeneity Index and *N* is the total sample size. The transformation is linear, so it preserves the ordering and shape of the distribution, and it keeps the likelihood defined for observations at 0 and 1.

Random effects structure

The dataset had a hierarchical, cross-classified structure: individual pages generated multiple posts, and the same news links (content) were frequently shared across different pages. Ignoring this non-independence would lead to underestimated standard errors and inflated Type I error rates.

To account for this structure, we included random intercepts for both pages and content. We also included random slopes for Tox at both levels, so that the association between Tox and reaction homogeneity could differ across pages and news links instead of being assumed uniform.

S4 Appendix

Identity-based assessment of page bias

We examined all 117 pages in the analytical sample with at least 15 distinct links in their sharing history. We assigned Left, Center, or Right labels using identifiable politicians, parties or movements, explicit ideological self-descriptions, and the editorial identity of recognizable media outlets. Pages without an identifiable orientation were left uncoded (51/117). These labels were assigned without reference to *b_p*. We then categorized *b_p* as Left (*b_p* ≤ −0.3), Center (−0.3 < *b_p* < 0.3), or Right (*b_p* ≥ 0.3). Table 5 compares the two classifications.

Table 5. Identity-based labels compared with categories derived from *b_p* (66 coded pages). Rows are identity-based labels and columns are *b_p* categories.

	Left	Center	Right
Left	36	0	0
Center	1	8	2
Right	0	1	18

Agreement was 62/66 (93.9%). One page whose name partially matched a different political movement was left uncoded after its identity was checked. Coding it by the name-matching rule alone would have yielded 62/67 agreement (92.5%) and one Left–Right reversal. Media classifications require editorial judgments, and an audience-derived score for a media outlet need not match its editorial identity. This exercise therefore supports the face validity of *b_p* for identifiable political pages but does not establish it as an independent measure of every page’s ideology.

S5 Appendix

Reaction composition and negative-reaction share

Across the 4,584 analytical posts, Like accounted for 64.8% of all reactions. Considering the four reactions used to define positive and negative reactions (Like, Love, Sad, and Angry), 953 posts (20.8%) received only positive reactions, while 3,631 (79.2%) received reactions of both signs. No post received only negative reactions. Only 237 posts (5.2%) had a majority of negative reactions, and eight posts had equal positive and negative reaction counts. For the 4,347 posts (94.8%) with a negative-reaction share of at most 0.5,

$$RHI_i = 1 - 2f_i^-,$$

where f_i^- denotes the proportion of negative reactions on post i . The relationship between the negative-reaction share and RHI is therefore largely fixed by their definitions.

As an additional sensitivity analysis, we modeled the focal post's negative-reaction share directly. Both negative-share models described below retained the main model's signed bias and extremity terms, activity controls, topic predictors, and random-effects structure, while omitting *CRP*. Greater Bias Discrepancy was associated with a higher negative-reaction share in a mixed beta model (standardized $\beta = 0.147$, $p < 0.001$). We also estimated a beta-binomial model using the underlying negative and positive reaction counts. This model likewise showed that greater Bias Discrepancy was associated with a larger share of negative reactions ($\beta = 0.190$, $p < 0.001$).

Because the negative-reaction share and RHI are mathematically related for most posts, these models are a sensitivity analysis of the reaction pattern captured by RHI , not independent evidence for a separate outcome.

Model specification and sensitivity analyses

We assessed whether the association between Bias Discrepancy and reaction homogeneity was sensitive to the representation of political bias and to additional model specifications. Using the same posts, outcome, and remaining covariates, the original specification containing signed content and page bias had an AIC of -14989.3 . Adding content and page extremity improved model fit (AIC = -15014.6), and a likelihood-ratio test comparing this model with the original signed-bias specification was significant ($\chi^2(2) = 29.24$, $p < 0.001$). Replacing the continuous content and page bias terms with Left/Center/Right categories yielded an AIC of -14998.8 . Of these three alternatives, the specification containing both signed bias and absolute extremity provided the best fit. The main model reported in Table 3 additionally included total reactions and page links as activity controls (AIC = -15046.2).

We next assessed the sensitivity of the key associations using the main model as the reference. Table 6 reports standardized coefficients for Bias Discrepancy and Toxicity after omitting Content Reaction Positivity, restricting pages by the number of distinct bias-scored links in their sharing histories, restricting posts to the Twitter collection window (October 1–December 11, 2018), reconstructing page bias without the focal link, and varying the minimum-reaction cutoff.

For the sharing-history restrictions, we counted only distinct links with a content bias score, which are the links that contribute to b_p . The page-activity control retained its original definition based on all distinct shared links. In the leave-link-out analysis, we removed all posts by a page sharing the focal news link from that page's sharing history, then recalculated page bias, page extremity, and Bias Discrepancy together. This excluded 371 posts from pages with no remaining bias-scored history, leaving 4,213

posts. As in the original calculation, a remaining history with only center-leaning links was assigned $b_p = 0$. Other predictors retained their original definitions. The 25-reaction sample was rebuilt from the source data using the same preprocessing and metric-construction procedures before applying the lower minimum-reaction cutoff. In every fitted sample, continuous predictors were standardized and the boundary adjustment of RHI was recomputed using that sample’s size.

Table 6. Sensitivity analyses of the main model for the Reaction Homogeneity Index. Except for the stated modification, each model retains the fixed- and random-effects specification in Table 3. Continuous predictors are standardized within each fitted sample. Coefficients are on the logit scale and intervals are two-sided 95% Wald confidence intervals.

Specification	β_{Δ_b}	95% CI	β_{Tox}	95% CI	N
Main model	-0.173	[-0.211, -0.135]	-0.149	[-0.196, -0.102]	4,584
CRP omitted	-0.178	[-0.219, -0.138]	-0.133	[-0.180, -0.086]	4,584
Pages with ≥ 2 bias-scored links	-0.201	[-0.241, -0.160]	-0.162	[-0.211, -0.113]	4,213
Pages with ≥ 5 bias-scored links	-0.219	[-0.264, -0.175]	-0.170	[-0.226, -0.113]	3,405
Twitter-window posts only	-0.173	[-0.214, -0.132]	-0.144	[-0.192, -0.096]	4,099
Leave-link-out page bias	-0.192	[-0.234, -0.150]	-0.165	[-0.214, -0.116]	4,213
Minimum 25 reactions	-0.171	[-0.206, -0.135]	-0.133	[-0.179, -0.088]	5,774
Minimum 100 reactions	-0.167	[-0.209, -0.126]	-0.177	[-0.229, -0.126]	3,558

The main model and all seven sensitivity analyses returned an optimizer convergence code of zero and a positive-definite Hessian, with finite coefficient estimates and confidence intervals. Across all eight specifications, Bias Discrepancy and Toxicity had negative coefficients, with all 95% confidence intervals excluding zero. The Bias Discrepancy association persisted after removing the focal link from the page-bias calculation, after restricting the sample to posts within the Twitter collection window, and under alternative minimum-reaction cutoffs. These checks support the robustness of the reported association to these specific analytical choices. They do not remove the limitations of estimating page ideology from observed sharing behavior or establish a causal effect. Because standardization was performed within each sample, coefficient magnitudes reflect the predictor distributions in those samples.

Omitting *CRP* from the main model reduced marginal R^2 from 0.762 to 0.154, while conditional R^2 remained similar (0.977 versus 0.979). Marginal and conditional R^2 were calculated using the Nakagawa method implemented in `performance::r2_nakagawa()`, with the lognormal approximation. This indicates that *CRP* accounted for substantial fixed-effect variation, while the Bias Discrepancy association persisted when this same-link reaction covariate was omitted.

Categorical bias specification

As a sensitivity analysis, we replaced the signed content and page bias terms with Left, Center, and Right categories, defined as $b \leq -0.3$, $-0.3 < b < 0.3$, and $b \geq 0.3$, respectively. Center was the reference category. The model used the same posts, outcome, and other predictors as the signed-bias model used for the specification comparison, without the two activity controls. Table 7 reports the bias terms and key associations.

The categorical specification yielded the same central conclusion regarding Bias Discrepancy: greater discrepancy remained associated with lower reaction homogeneity ($\beta = -0.163$, $p < 0.001$). The categorical terms additionally showed higher RHI for right-leaning content and right-leaning pages relative to their respective center

Table 7. Categorical bias specification for the Reaction Homogeneity Index. Center is the reference category and continuous predictors are standardized.

Predictor	β	95% CI	p
Content: Left	-0.047	-0.169 to 0.075	0.451
Content: Right	0.244	0.079 to 0.408	0.004
Page: Left	0.074	-0.067 to 0.215	0.304
Page: Right	0.308	0.154 to 0.463	< 0.001
Bias Discrepancy (Δ_b)	-0.163	-0.201 to -0.125	< 0.001
Toxicity	-0.146	-0.194 to -0.098	< 0.001
Content Reaction Positivity	0.962	0.899 to 1.024	< 0.001
AIC	-14998.8		

categories, whereas the corresponding left-versus-center differences were not statistically significant.

S6 Appendix

Composition of Bias Discrepancy quartiles

Bias Discrepancy (Δ_b) was divided into four groups using cut points of 0.161, 0.313, and 0.498. The maximum observed value was 1.746. The groups had nearly equal numbers of posts by construction. Table 8 reports the numbers of distinct pages and links and the shares contributed by the largest pages.

Table 8. Composition and page concentration of Bias Discrepancy quartiles.

Discrepancy level	Posts	Pages	Links	Largest page	Top 10 pages
Very Low	1,148	427	361	4.4%	23.3%
Low	1,148	554	419	6.7%	24.1%
High	1,142	499	375	8.0%	27.0%
Very High	1,146	595	374	3.9%	19.1%

No group was dominated by a single page. The Very High discrepancy group had the most distinct pages and the smallest shares contributed by its largest page and top ten pages.

S7 Appendix

Estimation and validation of Twitter-derived content bias scores

This appendix summarizes the procedure used by [26] to estimate the Twitter-derived content bias scores (b_c) employed in our study. In the original study, user polarity was estimated on a weekly basis. Users with fewer than five tweets in a given week were excluded, resulting in 72,576 users in the Brazilian dataset. The hashtags involved were those present in each user’s tweets for that week, including the ones they retweeted, rather than hashtags attached to the news links.

Reading a retweet as a signal of the retweeter’s own alignment is an established basis for estimating political position from Twitter interaction data [16, 28], and it is how retweeting was used throughout the original study and in prior work on the same election [11].

Hashtags were labeled in two stages. The 100 most frequent hashtags were classified as left, neutral, right, or uncertain by three volunteers who were not authors of the original study and who worked independently of one another. Only the 64 hashtags on which all three agreed were retained (Fleiss’ $\kappa = 0.63$). The remaining hashtags were labeled by propagating these seeds through a co-occurrence network, in which a node

represents a hashtag and an edge joins two hashtags used in the same tweet. This procedure classified 57,487 hashtags (65.6% of the 87,620 observed). Repeating the propagation 100 times while randomly hiding 10% of the manual labels gave $\kappa = 0.84$, while hiding 20% gave $\kappa = 0.78$.

Each user’s polarity was then computed as the contrast between the right- and left-leaning hashtags they used, considering all their classified hashtags, with per-class weights to account for the uneven numbers of left, neutral, and right hashtags in the dataset. The score for a news link aggregated the polarity of the users who retweeted it. Retweets were first counted across 21 user-polarity bins, from -1.0 to $+1.0$ in increments of 0.1 . These counts were normalized across the dataset and within each link before being weighted by the corresponding polarity values to obtain the content bias score. The resulting score is interpreted on the same scale as user polarity [26]. Thus, hashtags did not characterize the news links directly. Instead, they provided the initial ideological signal used to estimate user polarity, from which the content bias score was derived.

The original study also evaluated potential circularity in this procedure using synthetic null models. The correlations induced by the construction of the metric were weak relative to those observed in the empirical data, providing evidence that circularity had a limited impact on the reported associations [26].